TOKEN BALANCING, UPDATE SPARSITY, AND SUPERVISION DENSITY IN MULTI-TEACHER ON-POLICY DISTILLATION

Anonymous authors
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ABSTRACT

Multi-teacher on-policy distillation (MOPD) aims to consolidate specialized capabilities in a single student. We examine three aspects of its optimization: token balancing, update sparsity, and supervision density. Starting from the MOPD learning objective, we first show how loss averaging determines both domain weights and the relative weight of long and short responses. A finite-batch gradient decomposition separates these effects and motivates matched experiments on capability balance. Second, we investigate whether parameter changes concentrate on small, overlapping subsets when different teachers act on the same MOPD student, and whether more different teacher distributions correspond to more different subsets. We measure optimizer steps and cumulative parameter changes separately, using raw gradients to help explain the observed updates. Third, we compare sampled-token policy gradients with teacher top-k supervision in single-teacher and joint training, using full-vocabulary reverse-KL updates at shared prefixes as a local reference. The planned experiments pair update concentration with per-domain capability and account for sampling variability, training exposure, and compute. Each question is developed together with its experimental design, without presuming that sparse updates, low teacher overlap, or sampled supervision improve capability integration. Empirical outcomes remain to be measured.

1 INTRODUCTION

Multi-teacher on-policy distillation (MOPD) transfers domain specialists into one student by scoring student-generated responses with their assigned teachers (Ma et al., 2026). The objective is to integrate their capabilities, but equal numbers of prompts from each domain need not give those domains equal influence on training. Open-MOPD identifies capability imbalance and studies token-share weighting, changing distillation gaps, and stale student-dependent rewards (Gao et al., 2026). These findings motivate examining how the objective weights each domain and which student parameters its teacher changes most.

Prior work reports sparse cumulative parameter changes in RL (Mukherjee et al., 2025) and OPD (Yu et al., 2026; Shen et al., 2026). We use update sparsity to describe concentration of actual parameter changes, distinguishing individual optimizer steps from cumulative changes over training. We ask how multiple teachers affect a common student during MOPD, and use gradients as an additional measurement to separate the current loss signal from optimizer history. Independently trained students provide a reference, but their endpoints do not identify teacher contributions within the joint student.

After introducing the MOPD learning objective in Section 2, we develop three studies and place each experiment alongside its corresponding question:

1. **Token balancing (Section 3).** We derive how global-token, domain-token, and domain-response averaging weight domains and response lengths, then compare their effects on capability balance. The analysis explains existing loss reductions without introducing a new allocation algorithm.

2. **Update sparsity (Section 4).** We measure the concentration of optimizer steps and cumulative parameter changes in single-teacher and joint OPD, compare the parameters most changed by each teacher at the same MOPD checkpoint, and relate their overlap to teacher distribution differences. Raw gradients help interpret these updates.
3. **Supervision density (Section 5).** We compare sampled-token PG and teacher top-k training in both settings, with full-vocabulary updates on common prefixes as a local reference. We test whether update sparsity differs materially and report capability alongside it.

Shared parameter locations can receive compatible or opposing changes, and sampled-token feedback still depends on all logits through the softmax normalizer. Neither low overlap nor limited vocabulary coverage therefore guarantees sparse parameter updates or better integration. **[TODO: Replace the prospective empirical contributions with observed findings and their measured scope once the experiments are complete.]**

2 MOPD Learning Objective

We follow the notation and loss definitions of Ma et al. (2026), Section 3.2. The student policy is π_θ and the frozen teacher for domain d is π_{ϕ_d}. A prompt x is drawn from the training mixture and the student generates a response $y \sim \pi_\theta(\cdot \mid x)$. At position t, both models condition on the same prefix $(x, y_{<t})$. Following MOPD, we abbreviate these conditional probabilities as $\pi_\theta(v)$ and $\pi_{\phi_d}(v)$, where v is a vocabulary token. Teachers and student share the vocabulary and tokenizer.

Reverse-KL objective. MOPD minimizes the student–teacher reverse KL at each response position and averages over the response length $|y|$:

$$\mathcal{L}_{\text{rev-KL}} = \mathbb{E}_{x,\, y\sim\pi_\theta}\left[\frac{1}{|y|}\sum_{t=1}^{|y|}\sum_{v}\pi_\theta(v)\log\frac{\pi_\theta(v)}{\pi_{\phi_d}(v)}\right]. \tag{1}$$

Each prompt is scored by its assigned teacher. The factor $1/|y|$ gives each response a mean token loss; it does not give every token in the entire training batch equal weight. This distinction motivates Section 3.

Sampled-token policy gradient. MOPD assigns each generated token a detached log-probability advantage, $\hat{A}_{\text{MOPD},t} = \text{sg}[\log \pi_{\phi_d}(y_t) - \log \pi_\theta(y_t)]$, where sg stops gradients. The clipped value is $\hat{A}^{\text{clip}}_{\text{MOPD},t} = \text{clip}(\hat{A}_{\text{MOPD},t}, -A_{\max}, A_{\max})$, and the policy-gradient (PG) loss is

$$\mathcal{L}^{\text{PG}}_{\text{MOPD}}(\theta) = -\mathbb{E}_{x,\, y\sim\pi_\theta}\left[\frac{1}{|y|}\sum_{t=1}^{|y|}\hat{A}^{\text{clip}}_{\text{MOPD},t}\log\pi_\theta(y_t)\right]. \tag{2}$$

This loss uses the sampled token y_t at each position. Clipping is part of the practical recipe and must be recorded when comparing losses.

Teacher top-k distillation. Let $\mathcal{T}_t^d = \text{TopK}_k(\pi_{\phi_d}(\cdot \mid x, y_{<t}))$ be the teacher's k most probable tokens. MOPD instead evaluates

$$\mathcal{L}^{\text{TopK}}_{\text{MOPD}}(\theta) = \mathbb{E}_{x,\, y\sim\pi_\theta}\left[\frac{1}{|y|}\sum_{t=1}^{|y|}\sum_{v\in\mathcal{T}_t^d}\left(\pi_\theta(v)\log\frac{\pi_\theta(v)}{\pi_{\phi_d}(v)} - \pi_\theta(v) + \pi_{\phi_d}(v)\right)\right]. \tag{3}$$

These are full-vocabulary probabilities, without renormalization over the selected tokens. The correction $-\pi_\theta(v)+\pi_{\phi_d}(v)$ makes each selected-token term minimal when the two probabilities match; it does not recover the omitted vocabulary terms. With the entire vocabulary included, the correction sums to zero and the loss reduces to Equation 1.

Gradients measured in this paper. We differentiate the chosen loss on fixed student-generated responses, holding prefixes, masks, teachers, and detached advantages fixed. These local training gradients do not include derivatives through the generation of the responses or their lengths. The unclipped sampled-token gradient matches the full reverse-KL gradient in expectation at a fixed prefix; clipping, top-k truncation, and trajectory-dependent weighting need separate treatment (Section 5).

3 Token Balancing and Response Length

3.1 How does loss averaging change domain weights?

The MOPD objective in Section 2 averages token losses within each response. Open-MOPD analyzes a different implementation: averaging all response tokens together gives instruction following approximately 1% of the token weight despite supplying 20% of the prompts (Gao et al., 2026). Length imbalance therefore depends on the averaging rule; it is not intrinsic to the response-normalized MOPD objective.

For response y_i, write its mean token loss as $\ell_i = |y_i|^{-1} \sum_t \ell_{i,t}$, using any of the per-position losses in Section 2. Let $\mathcal{B}_d$ index the responses from domain d in the batch. The intended domain weights are $\lambda_d \geq 0$, with $\sum_d \lambda_d = 1$. Table 1 compares three averaging rules using only these quantities.

Averaging rule	Batch loss
All tokens together	$\left(\sum_i \lvert y_i\rvert \ell_i\right) / \left(\sum_i \lvert y_i\rvert\right)$
Tokens within each domain	$\sum_d \lambda_d \left(\sum_{i \in \mathcal{B}_d} \lvert y_i\rvert \ell_i\right) / \left(\sum_{i \in \mathcal{B}_d} \lvert y_i\rvert\right)$
Responses within each domain	$\sum_d \lambda_d \lvert\mathcal{B}_d\rvert^{-1} \sum_{i \in \mathcal{B}_d} \ell_i$

Table 1: Loss averaging on the same rollout batch. Each domain has at least one nonempty response. The last rule recovers an ordinary response mean when λ_d equals the domain's share of responses.

With equal numbers of prompts, a domain producing responses ten times as long receives ten times the weight under the first rule. The second rule fixes each domain's total weight to λ_d, as in Open-MOPD's token-share balancing, but longer responses still count more within that domain. The third rule also gives responses equal weight inside each domain. Thus domain balancing and response-length normalization address different effects.

The same distinction holds for gradients. Writing the response gradient as $g_i = \nabla_\theta \ell_i$, the token-weighted gradient within a domain equals its mean response gradient plus the covariance between response length and g_i, divided by mean response length. Appendix B gives the exact fixed-batch identity and its short proof. This is a weighted-average identity, not a new optimizer. Equal domain weights do not imply equal gradient magnitudes or compatible gradient directions.

3.2 Experiment: compare averaging rules on identical batches and online

Shared setting. All three studies use a Qwen3-1.7B student on mathematics, code, instruction following, and science. Math and instruction following have domain-specific Qwen3-1.7B RL teachers; code and science currently share one Qwen3-4B teacher. The reference run uses equal domain prompt quotas and weights, 64 responses per update, one update per fresh rollout batch, and a $4{,}096$-token response cap. The initial budget is at most 500 updates per run, subject to a throughput pilot. Appendix A records the shared setup and evaluation protocol. The experiments below are planned; measured results are not yet available.

Fixed-batch comparison. At early, intermediate, and late checkpoints, apply all three rules to the same responses and corrected teacher top-64 losses. Record domain token shares, response lengths, gradient norms and directions before clipping, and the length–gradient covariance. Check the decomposition and evaluate each proposed step on a common held-out response bank. Accumulate the correct sums and denominators across micro-batches and workers: averaging unequal micro-batch means can change the intended loss. An unequal-quota check first matches λ_d to prompt shares, then tests uniform weights separately, so changes in the domain prior are distinguishable from length weighting. **[TODO: Loss-reduction audit and gradient measurements.]**

Online comparison. Train three branches differing only in the averaging rule. They share initialization, teachers, prompt schedule, length limits, and optimizer settings; each generates responses from its own evolving student. The response-averaged top-64 branch is reused in Sections 4 and 5. Report per-domain capability, macro-average, and worst-domain change against both updates and generated tokens, with GPU time reported separately. A supporting $1{,}024$ versus $4{,}096$ cap comparison records truncation and answer-completion rates. Shortening responses changes the available prefixes and may remove solution endings, so it is a separate intervention from loss reweighting. Neither the algebra nor the cap comparison guarantees a capability improvement. **[TODO: Per-domain capability curves for the three averaging rules.]**

4 Update Sparsity and Teacher Overlap

We ask whether OPD parameter updates concentrate on a small fraction of parameters, whether different teachers affect the same parameters within one MOPD student, and whether teacher distribution differences predict this overlap. Each question includes its planned experiment; no outcome is assumed.

4.1 Does update sparsity persist under multiple teachers?

Prior work reports concentrated parameter changes in RL fine-tuning (Mukherjee et al., 2025) and OPD (Yu et al., 2026; Shen et al., 2026). We measure cumulative parameter changes $\Delta = \theta - \theta_0$ and actual optimizer steps $u = \theta_{\text{after}} - \theta_{\text{before}}$ separately. For each, we report the fraction of parameters with absolute change at most a threshold τ, together with the fraction of squared norm carried by the largest-magnitude ρ fraction of parameters. We sweep τ and ρ: "update sparsity" here means near-zero parameter changes or concentrated changes, not necessarily exact zeros. Per-step and cumulative sparsity can differ because successive updates can cancel.

Experiment. Using the representative single-teacher and joint runs from Section 5, record cumulative changes at early, intermediate, and final checkpoints and actual optimizer steps around those checkpoints and initialization. Report curves against both domain exposure and compute. Measure differences using FP32 master weights when available and report exported BF16 checkpoint differences separately to identify rounding-induced zeros; converting BF16 values to FP32 cannot recover lost changes. Threshold curves, norms, and energy concentration jointly show whether apparent sparsity reflects small overall scale or concentration on particular parameters.

As a secondary diagnostic, compare each domain's raw gradient $g_d = \nabla_\theta \mathcal{L}_d$ and the combined MOPD gradient under the same loss reduction, where $\mathcal{L}_d$ is that domain's mean loss. Match diagnostic response counts per domain across single-teacher and joint students, report valid-token counts, and check sensitivity to batch size; record the combined gradient's full batch size separately. Repeat over fresh student-rollout batches, record accumulation denominators, and accumulate gradients in FP32 before clipping or optimizer processing. Apply the same threshold, norm, and concentration summaries to gradients, keeping their scales separate from steps and cumulative changes. AdamW momentum and weight decay can move parameters with a zero current gradient, so gradient sparsity alone cannot establish update sparsity.

4.2 Do teachers affect the same parameters in one student?

At a fixed MOPD checkpoint θ, compute a proposed optimizer step for each routed domain using its teacher π_{ϕ_d} and student-generated responses. Each calculation starts from an identical copy of the student and optimizer state, without advancing training. These steps measure the local effect of applying each teacher's loss; they do not identify fixed, isolated subnetworks. To compare locations fairly, first select the same fraction of parameters for every teacher, ranked by absolute proposed change. For domains d and e,

$$\text{overlap}(d, e) = \frac{\text{number of parameters selected by both teachers}}{\text{number of parameters selected by either teacher}}. \tag{4}$$

Zero means no shared selected parameters; one means identical selections. The corresponding difference is one minus this overlap. We also report absolute-threshold selections, which retain information about update scale as well as location.

Experiment. Use the matched diagnostic response counts and loss reduction above. Repeat over diagnostic batches and report overlap matrices across checkpoints, with concentration curves alongside them. Compare global selections and selections made separately within each layer against random selections with matching counts, including per-layer results. Signed step alignment provides additional context: shared parameters can receive compatible or opposing updates, so lower overlap is not assumed better.

Separately compare raw-gradient overlap and alignment and use the zero-gradient optimizer control in Appendix C.1 to check whether shared momentum or decay dominates proposed-step overlap. These steps do not decompose the joint AdamW step or past changes by teacher. Independently trained single-teacher endpoint changes provide a reference only: those students follow different trajectories and cannot identify teacher contributions inside the joint student. Appendix C.1 specifies recording and threshold conventions.

4.3 Do more different teachers affect more different parameters?

We measure the difference between teachers π_{ϕ_d} and π_{ϕ_e} by their Jensen–Shannon divergence, a symmetric, bounded measure of how differently they distribute probability over next tokens. Every teacher scores the same cached, held-out student prefixes, with common domain weights. We also record each teacher's difference from the current student π_θ and its gradient and step norms. Appendix C.2 gives the distribution-distance formula.

Experiment. At each checkpoint, plot teacher distribution difference against the parameter-selection difference from the routed proposed steps above. A smaller control recomputes teacher steps and gradients on identical prefixes, testing whether an association persists when the response contexts also match. For PG, pair the same freshly sampled next tokens and fixed prefix weights across teachers, repeating the paired draws across batches. Available intermediate teacher checkpoints can add variation without training new teachers. Code and science share one Qwen3-4B teacher, so their routed comparison reflects context differences rather than two independent teacher models.

Show individual pair trajectories and run-level uncertainty. Pairs that share a teacher or training lineage, and repeated checkpoints, are dependent observations. Relate teacher–student differences to update concentration and observed transfer as additional analyses. A small teacher pool supports a setting-specific association, not a universal monotone relationship; larger, smaller, or unchanged parameter overlap are all valid outcomes.

5 Does Supervision Density Change Update Sparsity?

Our third study asks whether vocabulary-level supervision changes how strongly parameter updates concentrate on a small fraction of model parameters. Sampled-token PG uses feedback for the generated token, full-vocabulary reverse KL includes every vocabulary alternative, and teacher top-k provides a practical intermediate choice (Section 2). This concerns vocabulary coverage at each prefix, rather than the number of supervised response positions. Sampled-token feedback does not imply a sparse parameter gradient: the sampled token's log probability depends on all logits through the softmax normalizer. Concentrated OPD parameter changes have also been observed with dense feedback (Yu et al., 2026). We therefore test for a difference without assuming that PG produces sparser updates or better capability integration.

5.1 Matched single-teacher and joint experiments

We select one representative single-teacher setting before inspecting results and compare PG with corrected teacher top-64 in that setting and in joint MOPD (Table 2). These four online configurations use the losses in Equations 2 and 3; the joint top-64 run also anchors Section 3. All use domain-response averaging and match initialization, teachers, optimizer, loss masks, response cap, prompt schedule, and update budget within each setting. The top-64 loss uses probabilities normalized over the full vocabulary, without renormalizing within the selected tokens; we record any clipping. The shared model and training setup appears in Section 3 and Appendix A.

At the initial, early, intermediate, and final checkpoints, we evaluate all three losses on a small shared bank of student-generated prefixes. Each comparison holds the student, prefix weights, and optimizer state fixed; local PG uses fresh token samples at those prefixes. Full-vocabulary reverse-

Setting	Online losses	Common-prefix reference
Single teacher	PG, teacher top-64	Full-vocabulary reverse KL
Joint MOPD	PG, teacher top-64	Full-vocabulary reverse KL

Table 2: Supervision-density experiments. Four online configurations share full-vocabulary local measurements at matched checkpoints.

KL proposed steps provide the local reference in both settings, with raw gradients as a diagnostic; these local measurements cannot establish the cumulative changes of a full-vocabulary online run, which is conditional on measured resource feasibility. Online PG and top-64 branches generate their own on-policy responses, so their trajectories compare practical loss choices rather than isolating vocabulary coverage as the only causal difference.

5.2 Update concentration and learning outcomes

Our primary measurements are actual online steps and cumulative parameter changes from initialization, reported separately. Following Section 4, we report threshold curves, norms, and the fraction of squared change carried by the largest coordinates. We use FP32 master-weight differences when available, with exported BF16 checkpoint differences reported separately. Local comparisons use proposed steps from copies of the same optimizer state; repeated batches on the common prefix bank quantify sampling variability. Raw gradients before clipping provide secondary concentration measurements. Their sparsity need not equal update sparsity because momentum, clipping, weight decay, rounding, and cancellation across steps can change which parameters move.

For the joint runs, we also compare which parameters the teachers' proposed steps select, with raw-gradient overlap as a diagnostic, using Section 4's overlap measurement. Held-out per-domain capability, the macro-average, and worst-domain change accompany the sparsity results. Curves show generated-token exposure and GPU hours separately, with response counts, updates, and teacher-scoring and diagnostic overhead recorded.

5.3 Interpretation and decision rule

At a fixed prefix, fresh unclipped sampled-token PG with a detached log-ratio coefficient has the same expected gradient as full-vocabulary reverse KL (Yu et al., 2026; Oh et al., 2026). This identity holds with fixed prefix weights; weighting an original rollout token by its eventual response length need not preserve it. Clipping and top-k truncation also change the comparison, and equality in expectation does not equate sampled gradients, nonlinear optimizer steps, or training trajectories.

We report paired-seed effect estimates and confidence intervals against a material difference margin specified before examining results. One-seed runs remain exploratory; prompt resampling measures evaluation uncertainty, not training-seed variation. Intervals entirely within the margin support no material sparsity difference in the tested regime; intervals spanning negligible and material effects are inconclusive. Only a stable, meaningful gap motivates the optional sampling-variance and truncation controls in Appendix C.3.

[TODO: PG/top-64 optimizer steps, cumulative changes, and capability; full-vocabulary local references and raw-gradient diagnostics.]

6 Related Work

Capability integration and task weighting. MOPD consolidates routed domain teachers in policy space (Ma et al., 2026). Open-MOPD diagnoses token-share and reward-related imbalances (Gao et al., 2026). Our first study separates cross-domain weighting from within-domain response-length weighting in existing loss reductions. Prior work on multi-task loss weighting and gradient conflict also motivates distinguishing weight balance from joint capability gains (Chen et al., 2018; Yu et al., 2020).

Sparse parameter changes and teacher differences. Mukherjee et al. (2025) report sparse cumulative parameter changes in RL and repeated changes in similar parameter subsets across settings. Yu et al. (2026) establish related OPD signatures, including overlap when varying teachers and experiments that restrict training to selected parameters. Shen et al. (2026) examine update subspaces and response-position thinning. These results motivate our study but do not establish sparsity of instantaneous gradients. Teacher compatibility also affects distillation success (Li et al., 2026; Ma et al., 2026). We measure single-step and cumulative parameter changes separately and ask whether teachers with more different distributions change more different subsets of a shared MOPD student. Raw gradients help explain the role of the current loss signal.

Vocabulary coverage and gradient estimation. MOPD provides sampled-token and corrected teacher-top-k objectives (Ma et al., 2026). Variance-reduced sampled OPD (Oh et al., 2026), exact-head plus sampled-tail estimation (Zhang & Ba, 2026), and tail-aware losses (Huang & Wei, 2026) motivate separating vocabulary coverage from the choice of objective and estimator. Our loss study compares update concentration under practical PG and top-k training, with full-vocabulary local updates and their gradients in both single-teacher and joint settings. Established estimator controls are reserved for interpreting a stable difference; we do not propose a new estimator.

7 Discussion

The three studies connect loss weighting, parameter-update concentration in a shared MOPD student, and vocabulary-level supervision. Their interpretation depends on distinguishing raw gradients from optimizer steps and cumulative parameter changes. A concentrated cumulative change need not imply that each gradient or optimizer step was similarly concentrated. Overlapping update locations can also contain either shared or opposing corrections.

The PG/top-k comparison tests for a material difference rather than assuming an advantage for sampled supervision. A resolved absence of such a difference is informative within the tested regime; a stable gap can motivate further analysis. Likewise, teacher distribution differences may or may not predict which parameters change most. Capability measurements determine whether these optimization patterns accompany useful integration.

[TODO: State the supported observations, including null relationships, and their model, task, teacher, loss, and measurement scope after completing the experiments.]

AI use statement

Generative AI tools assisted with manuscript organization, language, mathematical analysis, and experimental design. The authors remain responsible for reviewing claims, proofs, code, and results.

Ethics statement

The proposed study uses existing language models and reasoning and instruction-following datasets. Model and data biases and automated verifier errors can affect capability comparisons. Data provenance, invalid outputs, and evaluator failures should accompany the results.

Reproducibility statement

Sections 3–5 present the three studies together with their experimental designs. Appendix A records the shared setup, and Appendix C.1 specifies the parameter-change and supporting gradient measurements. The repository currently contains the manuscript, analytical calculations, and study specifications; measured outcomes for these experiments are not yet available. **[TODO: Release runnable training and diagnostics, frozen configurations, source data for figures, per-seed outputs, exact checkpoints, and measured resource use before claiming empirical reproducibility.]**

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A SHARED EXPERIMENTAL DETAILS

Sections 3–5 contain the experiments; this appendix records shared resource and evaluation details.

Training setup and run reuse. Section 3 specifies the student, domains, and three teacher weight sets. The 64-response update comprises 16 micro-batches of four responses, with equal domain quotas, one optimizer update per fresh rollout batch, and a 4,096-token cap. The initial budget is at most 500 updates per run, subject to a throughput pilot on a 96-GB learner GPU and 48-GB rollout/teacher GPUs. Exact model revisions, optimizer and decoding settings, data splits, and evaluation versions remain to be frozen before execution.

A representative single-teacher setting is chosen before inspecting results. PG and corrected teacher top-64 in the single-teacher and joint settings give four configurations. The joint top-64 configuration with domain-response averaging is also the reference for the three-way normalization study, which adds two averaging rules. Thus six training configurations cover the core comparisons, with checkpoints reused for the update, teacher-overlap, and full-vocabulary measurements. Additional teacher settings, the response-cap comparison, and repeated seeds require separate resource accounting. The manuscript does not yet report measured outcomes for these studies.

Capability evaluation. Use MATH-500 greedy pass@1, a frozen LiveCodeBench pass@1 slice, IFBench strict accuracy, and GPQA-Diamond average@4. Record evaluator, model, and data revisions and fix decoding within each comparison. Report raw per-domain scores, their macro-average, and the worst-domain change from initialization. Independently trained OPD students, where available, are references at matched domain exposure, not upper bounds or guarantees of jointly attainable capability. A lack of single-teacher transfer is not itself evidence of multi-teacher interference. Loss on a common held-out response bank and loss on fresh student responses are reported separately; a lower mean KL alone does not establish balanced capability integration.

Compute and uncertainty. Record generated tokens, domain response counts, optimizer updates, and GPU hours separately, including teacher-scoring and diagnostic overhead. Key online comparisons use repeated paired training seeds when feasible; initial one-seed runs are exploratory. Prompt bootstrap estimates conditional evaluation uncertainty, not training-seed variation. Report effect sizes, confidence intervals, and the range of differences the experiment can resolve. Teacher pairs sharing weights or training lineage, and repeated checkpoints, are not independent observations. No sparsity advantage or capability improvement is claimed before measurement.

B LENGTH-WEIGHTING DERIVATION

Fix a rollout batch, its valid-token masks, and the domain weights λ_d. Any PG advantages are detached. Use the response mean ℓ_i and response gradient $g_i = \nabla_\theta \ell_i$ from Section 3. For $n_d = |\mathcal{B}_d|$, define mean response length $\bar{T}_d = n_d^{-1} \sum_{i\in\mathcal{B}_d} |y_i|$ and mean response gradient $\bar{g}_d = n_d^{-1} \sum_{i\in\mathcal{B}_d} g_i$. The empirical length–gradient covariance uses the same $1/n_d$ normalization:

$$\mathrm{Cov}_d(|y|, g) = \frac{1}{n_d} \sum_{i\in\mathcal{B}_d} (|y_i| - \bar{T}_d)(g_i - \bar{g}_d), \qquad \frac{\sum_{i\in\mathcal{B}_d} |y_i| g_i}{\sum_{i\in\mathcal{B}_d} |y_i|} = \bar{g}_d + \frac{\mathrm{Cov}_d(|y|, g)}{\bar{T}_d}. \tag{5}$$

To verify the identity, expand the covariance as $n_d^{-1} \sum_i |y_i| g_i - \bar{T}_d \bar{g}_d$ and divide by $\bar{T}_d$. Global token averaging combines the right-hand side using each domain's observed token share; domain token averaging uses λ_d instead. Domain response averaging gives $\sum_d \lambda_d \bar{g}_d$. Thus the change in the domain coefficient and the within-domain length–gradient covariance are distinct, even with identical rollouts. Multiplying a domain's globally averaged token terms by its target weight divided by its observed token share gives the domain-token rule exactly.

These are finite-batch identities. The expectation of a sample ratio is generally not the ratio of population expectations; replacing it with $\mathbb{E}[|y|g]/\mathbb{E}[|y|]$ requires an appropriate large-batch limit. The identities also do not differentiate through the response sampling distribution, masks, or response lengths.

C MEASUREMENT DETAILS AND OPTIONAL FOLLOW-UP

C.1 TEACHER-CONDITIONED MASKS AT A COMMON JOINT CHECKPOINT

The diagnostic record contains the joint student checkpoint, optimizer state, teacher identity, routed domain, prompt IDs, sampled response and mask, loss reduction, and gradient accumulation denominator. Each teacher-conditioned gradient and proposed step is computed from the same restored student state. They describe the parameters most affected by a teacher at that checkpoint, not a unique historical attribution of the cumulative joint delta.

Record raw and clipped gradients separately. Retain the actual optimizer step and its momentum and decay settings. A zero-gradient optimizer branch can identify motion caused by stored momentum or weight decay when those components dominate an overlap comparison. Prefer FP32 master-weight differences for cumulative updates when available; converting already rounded BF16 checkpoints to

FP32 does not recover unrecorded updates. Record exclusions, thresholds, relative scales, tie rules, and per-layer selected-parameter counts.

Comparisons selecting an equal fraction of parameters accompany absolute-threshold selections. A stratified random baseline preserves layer and source-weight-magnitude bins when feasible. Degenerate or tie-dominated gradients are marked; ranking a zero vector does not define a meaningful learned subnetwork. If both selections are empty, report overlap as undefined and exclude that comparison from averages. If only one is empty, overlap is zero.

C.2 Teacher distribution difference

For teachers d and e at the same prefix, the Jensen–Shannon divergence is

$$\mathrm{JS}(\pi_{\phi_d}, \pi_{\phi_e}) = \tfrac{1}{2} D_{\mathrm{KL}}(\pi_{\phi_d} \| m) + \tfrac{1}{2} D_{\mathrm{KL}}(\pi_{\phi_e} \| m), \qquad m = \tfrac{1}{2}(\pi_{\phi_d} + \pi_{\phi_e}). \tag{6}$$

It lies between zero and $\log 2$ with natural logarithms and is zero when the two next-token distributions agree. Average this quantity across the same prefix bank and domain weights for every pair. Replacing one teacher distribution with π_θ gives the teacher–student comparison. Truncated estimates must report their retained probability mass and must not be presented as exact full-vocabulary divergences.

C.3 Optional analysis if the loss comparison shows a stable difference

If a stable difference warrants further analysis, resample next actions on a fixed student-prefix bank and compare sample counts, an action-independent detached baseline, and the exact full-vocabulary gradient. These controls test whether the effect tracks sampling variance. A retained-head plus sampled-tail estimator can distinguish truncation from the full reverse-KL reference (Oh et al., 2026; Zhang & Ba, 2026; Yu et al., 2026).

The independent action draws do not extend the response; prefix selection and weights remain fixed. Original rollout tokens with eventual response-length weights need not satisfy the same sampling identity. Forward KL, teacher versus student top-k, or within-set renormalization are objective changes and are labeled separately.

C.4 An optional amplitude calibration for teacher distance

A simple artificial target can help distinguish gradient magnitude from parameter selection. At a fixed prefix $(x, y_{<t})$, write $\pi_0 = \pi_{\theta_0}$ for the reference student and freeze the teacher π_{ϕ_d}. Define the normalized interpolated distribution

$$\pi_\alpha(v) = \frac{\pi_0(v)^{1-\alpha} \pi_{\phi_d}(v)^\alpha}{\sum_{v'} \pi_0(v')^{1-\alpha} \pi_{\phi_d}(v')^\alpha}, \qquad \nabla_\theta D_{\mathrm{KL}}(\pi_\theta \| \pi_\alpha)\big|_{\theta_0} = \alpha \, \nabla_\theta D_{\mathrm{KL}}(\pi_\theta \| \pi_{\phi_d})\big|_{\theta_0}. \tag{7}$$

All probabilities condition on the same prefix. The log-ratio is an α-scaled original log-ratio plus a constant across vocabulary tokens; the constant vanishes because $\sum_v \nabla_\theta \pi_\theta(v) = 0$. For $\alpha > 0$, positive scaling preserves the ranking of raw-gradient magnitudes while possibly changing absolute-threshold sparsity. This artificial calibration does not replace real-teacher comparisons or imply matched downstream capability.

C.5 Reproducibility records

Record exact student, teacher, tokenizer, data, and evaluator revisions; teacher training lineages; prompt schedules and seeds; decoding, EOS handling, cap, invalid responses, and loss masks; learning-rate schedule, optimizer moments, precision, clipping, and decay; candidate sets, probability normalization, teacher/student retained mass, and advantage handling. Keep per-prompt outputs, source measurements for every plot, and actual time and memory costs.

Distinguish training seeds, diagnostic batch replicates, and evaluation resamples. Pairs sharing a teacher are dependent, and prompt-level uncertainty does not establish population-level teacher relationships. Historical synthetic GPAS calculations are not empirical evidence for these studies.